

Blockchain Development Tasks & Instructions

- **TASK 1: Professional LinkedIn Announcement**
 - **Objective:** Share your official internship acceptance on your professional network.
 - **Requirements:** Post a high-resolution copy of your official **Internship Offer Letter** on LinkedIn. Craft a compelling caption detailing your learning goals, tag **Progree**, and use hashtags: `#blockchain` `#solidity` `#web3` `#progree`.
- **TASK 2: Immutability Ledger Block Hashing Simulation**
 - **Objective:** Write a local cryptographic console model demonstrating core blockchain link security properties.
 - **Requirements:** Program a simplified block node engine using JavaScript or Python. Define block structural schemas holding transaction data, apply SHA-256 cryptographic hashing algorithms to generate block fingerprints, and write loops that link new blocks to previous hashes while verifying chain validity.
- **TASK 3: ERC-20 Standard Token Contract Implementation**
 - **Objective:** Code and compile a standard digital asset smart contract on an Ethereum virtual test network.
 - **Requirements:** Author a compliant ERC-20 token asset contract using Solidity inside the Remix IDE. Define total supply variables, build balance lookup maps, and configure transfer authorization rules. Compile and deploy the contract onto an Ethereum testnet environment (e.g., Sepolia), logging active deployment address variables.
- **TASK 4: Mini Project - Multi-Party Escrow Decentralized Application (DApp) Engine**
 - **Objective:** Architect an end-to-end Web3 decentralized funding application executing on-chain transactions.
 - **Requirements:** Author a smart contract coordinating multi-party funding balances. Enforce transaction conditional checks, protect functions against reentrancy attacks, and deploy automated refund pathways if project deadlines fail. Build a web frontend interface (using Web3.js or Ethers.js) that allows client crypto wallets (e.g., MetaMask) to connect and interact directly with the smart contract.